

MANDAL II, Norway

WMO: 014300

Lat: 58.0284N

Lon: 7.4527E

Elev: 453

StdP: 14.46

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	17.5	6.1	7.6	16.9	10.2	9.4	21.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	74.5	61.6	71.4	60.5	68.9	59.6	64.4	70.1	63.1	68.2	61.7	66.5	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.4	85.7	66.4	61.0	81.6	65.4	59.6	77.5	64.1	29.8	70.4	28.8	68.4	27.8	66.6	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	DB	11.7	78.4	5.3	2.3	7.9	80.0	4.8	81.4	1.8	82.7	-2.1	84.4
			WB	10.6	66.5	5.1	1.9	7.0	67.8	4.0	68.9	1.2	70.0	-2.5	71.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	33.4	33.6	38.1	43.8	51.3	57.4	61.7	60.0	55.6	47.9	41.2	36.2
	DBStd	11.46	6.92	6.94	5.84	5.19	6.48	3.84	4.18	3.00	3.84	4.73	6.26	8.35
	HDD50	2391	514	459	369	196	60	0	0	0	5	94	265	429
	HDD65	6682	979	879	834	635	427	233	121	156	282	529	713	894
	CDD50	1217	0	0	1	11	101	221	364	312	174	31	2	0
	CDD65	29	0	0	0	0	2	4	20	3	0	0	0	0
	CDH74	102	0	0	0	0	5	13	83	1	0	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	64.1	7.5	5.2	4.1	3.3	3.1	3.7	3.9	5.2	6.0	7.4	7.1	6.9
	PrecMax	91.2	16.7	13.4	9.4	5.6	6.3	8.3	9.6	9.5	11.9	13.3	19.9	13.4
	PrecMin	35.6	0.7	0.6	0.6	0.9	0.3	0.3	0.4	0.5	3.2	2.0	2.7	1.2
	PrecStd	12.0	3.5	3.4	1.9	1.3	1.7	2.1	2.2	2.1	2.0	3.2	3.8	3.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.1	49.5	58.1	62.5	74.7	75.8	80.0	72.8	68.3	60.4	54.4	50.7
		MCWB	44.6	43.8	47.8	49.0	59.8	61.1	63.9	63.6	60.1	55.4	51.8	48.2
	2%	DB	46.0	45.9	51.6	59.1	71.0	71.4	76.4	69.7	65.3	58.2	52.4	49.3
		MCWB	43.6	42.5	44.3	47.8	57.6	58.5	63.5	61.1	58.5	54.5	49.9	47.2
	5%	DB	44.7	43.9	48.4	55.8	66.8	68.4	72.9	68.0	63.6	56.5	51.1	48.2
		MCWB	42.5	41.0	42.3	46.7	56.3	57.3	62.1	60.4	57.9	53.1	48.6	46.1
	10%	DB	43.4	42.6	45.8	52.4	62.5	65.6	69.9	66.3	62.1	55.1	49.8	46.6
		MCWB	41.0	40.3	41.3	44.8	54.0	56.0	61.0	59.2	57.3	51.5	47.4	44.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	46.1	48.6	52.2	61.3	63.6	67.2	65.6	63.1	57.5	52.6	49.1
		MCDB	47.1	48.0	55.1	57.4	71.4	72.7	74.4	69.8	65.1	59.1	53.9	50.0
	2%	WB	44.0	43.5	45.8	49.6	59.2	60.9	65.2	63.8	61.4	55.8	50.6	47.7
		MCDB	45.6	44.7	50.0	56.0	68.6	67.8	72.1	67.5	63.5	57.7	52.0	48.9
	5%	WB	43.1	42.0	43.8	47.6	57.2	59.2	63.8	62.5	59.7	54.1	49.1	46.5
		MCDB	44.7	43.4	46.9	54.3	64.9	65.6	70.0	65.8	62.4	55.9	50.7	47.9
	10%	WB	41.3	40.5	42.2	45.6	54.9	57.5	62.4	61.2	58.0	52.6	47.9	44.7
		MCDB	43.1	42.4	45.0	51.1	61.2	63.3	68.5	64.4	60.9	54.6	49.8	46.3
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.8	10.5	11.7	13.0	12.8	12.1	11.4	10.6	9.3	7.7	8.1
		MCDBR	9.4	10.1	15.3	16.4	18.2	18.2	16.7	13.1	12.8	10.0	8.4	7.6
		MCWBR	8.9	7.9	9.8	9.0	9.1	9.3	8.2	7.2	8.0	7.4	7.6	7.8
	5% WB	MCDBR	9.4	9.3	12.8	14.3	15.9	14.9	13.6	9.9	9.8	8.2	7.9	8.1
MCWBR		9.2	8.3	8.6	8.5	8.7	8.8	7.4	7.1	7.8	7.4	7.8	8.5	
Clear-Sky Solar Irradiance	taub	0.300	0.326	0.342	0.379	0.368	0.356	0.372	0.375	0.355	0.335	0.307	0.279	
	taud	2.230	2.320	2.369	2.317	2.422	2.477	2.449	2.437	2.484	2.441	2.293	2.184	
	Ebn at Noon	170	224	256	263	275	280	272	262	250	220	167	139	
	Edh at Noon	16	23	28	34	33	32	32	30	25	20	15	12	
All-Sky Solar Radiation	RadAvg	108	299	729	1225	1657	1854	1676	1282	831	397	150	75	
	RadStd	22	61	76	136	194	168	178	103	84	60	31	10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-220	-292	N/A	N/A	

Nomenclature: See separate page